Showa Maintenance Co., Ltd. / SHOWAM ORGANIC PRODUCTS INC.

Organic Waste Treatment & Organic Fertilizer Production

Toward a Circular GX Agriculture Model in Angeles City

Presentation of Solid Waste Management Technology

Provincial Solid Waste Management Board Meeting — 2nd Quarter 2026

AGENDA

Today's Flow (about 30 minutes)

- 1. Company introduction
- 2. Background: two challenges — organic waste and fertilizer supply
- ▶ Video 1 How the rapid fermentation-drying technology works (˜3 min)
- 3. The technology and treatment process
- 4. Verifying the GHG reduction (Angeles City)
- ▶ Video 2 The circular GX agriculture model (˜3 min)
- 5. The FS project plan and roadmap
- 6. Summary

> We combine slide-based explanation with two short videos.

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Company Introduction

Bringing Japanese waste-management expertise to organic fertilizer in the Phili ppines

COMPANY

Circulating organic resources across two bases

Showa Maintenance Co., Ltd. (Japan)

- Founded 1991 in Toyohashi, Aichi Prefecture
- An SME engaged in industrial-waste collection, transport and treatment
- Hands-on expertise with organic waste — food and agricultural residues
- A mindset that treats waste as a resource, not just disposal

SHOWAM ORGANIC PRODUCTS INC. (Philippines)

- Based in Angeles City, Pampanga
- Manufacturing, selling and exporting organic fertilizer since 2018
- Organic fertilizer made from sugarcane residue
- Halal- and Organic-JAS-certified, with a record of export to Japan

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Background & Challenge

An urban waste problem and an agricultural fertilizer problem

BACKGROUND

Two social challenges exist at the same time

- The Philippines generates more than 61,000 tons of solid waste per day
- Under rules that restrict incineration, reliance on landfill continues
- Anaerobic decomposition of organics in landfills releases methane (CH4)
- In agriculture, import dependence, high prices and soil degradation persist
- One public market in Angeles City alone produces about 10 t/day of organic waste

> Our starting point: can we solve waste treatment and fertilizer supply together, as one?

City and farm — two challenges, one technology

Urban challenge (waste)

- Pressure on landfill capacity
- Methane emissions from landfills
- Public-health and environmental risks

Agricultural challenge (fertilizer)

- High prices and import dependence for chemical fertilizer
- Soil degradation and acidification
- A shortage of affordable, high-quality organic fertilizer

Video 1 How rapid fermentation-drying works

From organic waste to organic fertilizer in about 24 hours (˜3 min)

3

The Technology

The rapid fermentation-drying unit and its process

Key features of the rapid fermentation-drying unit

- Enclosed aerobic high-speed fermentation (bioreactor type)
- Uses indigenous aerobic microbes to treat mixed organic waste in about 24 hours
- Far faster than conventional composting (2–6 months)
- Enclosed design means low odor and a zero-wastewater design
- Capacity of about 10 t/day (matched to the market's waste volume)

> The output is limited to organic fertilizer. Plastics and other unsuitable materials are automatically separated within the process.

The process: from input to organic fertilizer

1 Input

Receive waste

Adjust moisture

Add & mix microbes

2 Ferment & dry

About 24 hours

Aerobic at 40–70°C

Continuous mixing, evaporation

3 Sorting (in-line)

Vibrating screen

Magnetic / air separation

Plastics auto-separated

4 Organic fertilizer

About 20% of input

~2 t/day, ~500 t/year

Recovered as quality fertilizer

Zero wastewater, low odor, no incineration. Plastics and other unsuitable items are automatically separated; only the organic fraction becomes fertilizer.

COMPARISON

Comparison with conventional methods

Item	This technology	Composting	BSF (insects)
Treatment time	About 24 hours	2–6 months	2–4 weeks
Odor	Very low (enclosed)	Strong (open-air)	Moderate
Quality stability	High (easy to control)	Low–mid (seasonal)	Depends on insects
Mixed waste	Yes (no pre-sorting)	Organics only	Food scraps only
Regulation (RA8749)	Compliant (non-incin.)	Compliant	Compliant
Use of output	Organic fertilizer	Compost	Feed / compost

Easy to deploy even where sorting infrastructure is limited; can be operated by about 2–3 local staff.

The output: high-quality organic fertilizer

- Rich in organic carbon, improving soil structure
- Slow-release nutrients (N-P-K) for sustained crop yield
- Improves water retention and aeration in tropical soils
- 100% organic soil-building, without reliance on chemicals
- Quality management aimed at Philippine fertilizer registration (FPA)

> The output is limited to organic fertilizer.

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Verifying GHG Reduction

An estimate for a public market in Angeles City

Assumptions (per IPCC 2006 Guidelines)

Parameter	Value	Source / basis
Waste treated	10 t/day	Measured at the public market (pre-survey)
Operating days	250 days/year	Assumes 5 days/week
Organic fraction	60–70%	ASEAN urban-waste statistics (IPCC)
CH_4 emission factor	0.06 t-CH_4/t	IPCC 2006 Vol.5
GWP (CH_4)	28	IPCC AR5 (100-year)
MCF correction	0.8	Assumes unmanaged landfill

Formula: 10 t/day × 250 days × organic fraction (60–70%) × 0.06 × 28 × 0.8 = about 2,016–2,352 t-CO_2e/year

Reduction from avoiding landfill methane

2,016–2,352

t-CO2e / year (landfill methane avoided)

Aerobic high-speed fermentation avoids methane generation from anaerobic decompo sition at its root.

From a single market only. A meaningful contribution to the Philippines' NDC (2030 target) is expected.

5 Circular GX Model & the FS Project

A feasibility study (FS) in Angeles City

Video 2 The circular GX agriculture model

The full loop that turns waste into a resource (˜3 min)

CIRCULAR MODEL

The circular GX agriculture model

Public market	Rapid ferment-dry	Organic fertilizer	Local farms	Back to market
Organic waste ~10 t/day	Resourced in ~24 hours	~2 t/day High quality	Supplied to farmers	Produce shipped via back-haul

Using the back-haul of vehicles that deliver to the market keeps logistics low-cost and low-carbon while widening the supply network.

What the FS project will verify

- The reality of waste volume, composition and collection
- Technical fit and operating performance in local conditions
- Quantified GHG reduction from avoiding landfill
- Quality, demand and marketability of the organic fertilizer
- The revenue model and business viability

FS = feasibility study. It sets the conditions needed to move to the next stage (demonstration, then commercial operation).

Business roadmap

Period	Phase	Main activities
2026–2027	FS study	Field survey, technical verification, GHG estimate, marketability
2027–2028	Pilot & fertilizer registration	Treatment trials, quality assessment, registration prep
2028–2029	Regional model verification	Finalize operations; horizontal-expansion plan to nearby cities
2029 →	Commercial & wider rollout	Commercialize in Angeles City → Luzon-wide and ASEAN

Alignment with national and local policy

- RA10068 (Organic Agriculture Act): aligned with promoting organic fertilizer
- RA8749 (Clean Air Act): a non-incineration treatment
- RA9003 (Solid Waste Management Act): reduces landfill reliance and adds recovery
- Philippines' NDC (2030): contributes to GHG goals by avoiding landfill methane

Expected outcomes

Resource circulation

- Fertilizer output: ˜500 t/year in year 1 → ˜2,000 t/year by year 5
- Waste treated: ˜2,500 t/year (10 t/day × 250 days)
- Lower landfill rate (toward zero waste)

Environment & community

- GHG reduction: ˜2,016–2,352 t-CO2e/year (landfill methane avoided)
- Gradually expanding organic-fertilizer supply to local farmers
- Less dependence on chemical fertilizer; less soil degradation

Summary

- Rapid fermentation-drying turns organic waste into fertilizer in about 24 hours
- Enclosed, low-odor, zero-wastewater, non-incineration; plastics auto-separated
- An estimated 2,016–2,352 t-CO2e/year reduction for Angeles City
- A circular GX model that solves waste treatment and fertilizer supply together
- The FS project will verify feasibility and impact quantitatively

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Waste into a resource. Value into agriculture.

A circular GX agriculture model via rapid fermentation-drying

Thank you for your attention.

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